

2.1

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

$$\bar{x}=1 \quad \bar{y}=2$$

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	-2
0	0	-1	1	-2	-2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

b.

$$b_1 = \bar{y} - b_2 \bar{x}$$

$$= 2 - b_2$$

$$= 2 - \frac{4}{5}$$

$$= \frac{6}{5}$$

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$= \frac{8}{10} = \frac{4}{5}$$

b_2 :
 x increase 1 unit
 y increase $\frac{4}{5}$ unit

$$b_1:$$

$$x = 0$$

$$y = \frac{6}{5}$$

$$c. \sum x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\sum x_i y_i = 12 + 4 + 3 + 1 + 0 = 18$$

$$\sum x_i^2 - N \bar{x}^2 = 15 - 5 \times 1 = 10 = \sum (x_i - \bar{x})^2$$

$$\sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \times 1 \times 2 = 8 = \sum (x_i - \bar{x})(y_i - \bar{y})$$

d. Use the least squares estimates from part (b) to compute the fitted values of $\hat{y}_i$ and complete the remainder of the table below. Put the sums in the last row.
 Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$\hat{y} = \frac{6}{5} + \frac{4}{5}x$$

$$= 1.2 + 0.8x$$

$$\bar{x} = 1 \quad \sum (x_i - \bar{x})^2 = 10$$

$$\bar{y} = 2 \quad \sum (y_i - \bar{y})^2 = 10$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$$

$$s_x^2 = \frac{\sum (x_i - \bar{x})^2}{N - 1} = \frac{10}{4} = 2.5$$

$$s_x = \sqrt{2.5} = \frac{\sqrt{10}}{2} = s_y$$

$$s_y^2 = \frac{\sum (y_i - \bar{y})^2}{N - 1} = \frac{10}{4} = 2.5$$

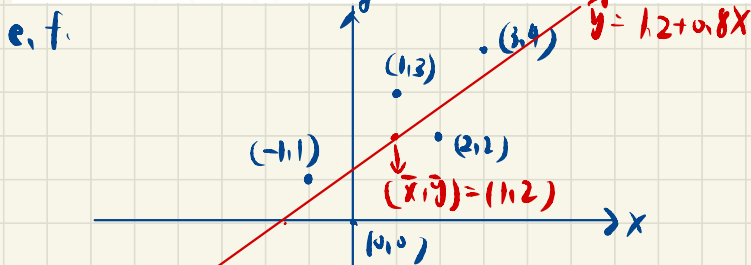
$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1} = \frac{8}{4} = 2$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5} \sqrt{2.5}} = 0.8$$

$$CV_x = 100 \times \left(\frac{s_x}{\bar{x}} \right) = 100 \times \left(\frac{1.5811}{1} \right) = 158.11\%$$

$$\text{median}(x) = 1$$

- On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.
- On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.
- Show that for these numerical values $\bar{y} = b_1 + b_2 \bar{x}$.
- Show that for these numerical values $\hat{y} = \bar{y}$, where $\hat{y} = \sum \hat{y}_i / N$.
- Compute $\hat{\sigma}^2$.
- Compute $\text{var}(b_2 | x)$ and $\text{se}(b_2)$.



g.

$$b_1 + b_2 \bar{x} = 1.2 + 0.8 \times 1 = 2$$

$$h. \sum \hat{y}_i = 10 \quad \bar{\hat{y}} = \frac{10}{5} = 2 = \bar{y}$$

$$i. \hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{n-2} = \frac{3.6}{3} = 1.2$$

$$j. \widehat{\text{Var}}(b_2 | x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$$

$$\text{se}(b_2) = \sqrt{0.12} \approx 0.34$$

HW0303_314707031

James

2026-03-09

```
load("~/Documents/Financial-Econometrics/Data/star5_small.rdata")
load("~/Documents/Financial-Econometrics/Data/cex5_small.rdata")
```

Problem 2.22

```
df <- star5_small
str(df)
```

```
## 'data.frame':    1200 obs. of  7 variables:
## $ aide          : int  0 0 1 1 0 1 0 0 0 1 ...
## $ mathscore     : int  528 454 463 449 478 536 444 528 478 459 ...
## $ readscore     : int  461 436 395 430 421 445 434 453 467 408 ...
## $ regular       : int  0 1 0 0 0 0 0 0 1 0 ...
## $ small         : int  1 0 0 0 1 0 1 1 0 0 ...
## $ tchexper      : int  13 8 8 3 15 11 12 6 11 8 ...
## $ totalscore    : int  989 890 858 879 899 981 878 981 945 867 ...
## - attr(*, "var.labels")= chr [1:7] "=1 if regular class with aide" "math score" "reading score" "=1
```

(a)

```
star_rs <- subset(df, aide == 0)
reg_a <- lm(totalscore ~ small, data = star_rs)
summary(reg_a)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = star_rs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -189.44  -50.44   -9.44   43.06  324.38
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 916.442      3.675 249.396 <2e-16 ***
## small      12.175      5.369   2.268  0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.59 on 773 degrees of freedom
## Multiple R-squared:  0.006608,    Adjusted R-squared:  0.005323
## F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

Interpretation of Estimates: $\text{TOTALSCORE}_i = 916.442 + 12.175 \times \text{SMALL}_i$

- Intercept ($\hat{\beta}_1 = 916.442$): The estimated average total aptitude score for students in regular-sized classes is 916.44.
- Coefficient for SMALL ($\hat{\beta}_2 = 12.175$): Students in small classes score, on average 12.175 points higher on their total aptitude score compared to students in regular-sized classes.
- Conclusion: The p-value for the small coefficient is 0.0236. Since this is less than 0.05 significance level, we reject the null hypothesis. We conclude that being in a small class has a statistically significant positive effect on learning (total aptitude scores).

(b)

```
reg_read <- lm(readscore ~ small, data = star_rs)
summary(reg_read)
```

```
##
## Call:
## lm(formula = readscore ~ small, data = star_rs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -74.665 -21.590  -2.665   16.410  194.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.488  290.760 < 2e-16 ***
## small        6.924       2.174   3.185  0.00151 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.2 on 773 degrees of freedom
## Multiple R-squared:  0.01295,    Adjusted R-squared:  0.01167
## F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
reg_math <- lm(mathscore ~ small, data = star_rs)
summary(reg_math)
```

```
##
## Call:
## lm(formula = mathscore ~ small, data = star_rs)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -144.777  -35.028   -5.028   29.223  142.223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.449  197.545  <2e-16 ***
## small        5.251      3.578   1.467   0.143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.71 on 773 degrees of freedom
## Multiple R-squared:  0.002778, Adjusted R-squared:  0.001488
## F-statistic: 2.153 on 1 and 773 DF, p-value: 0.1427
```

Observations and Differences:

- Reading Score: The estimated coefficient for small is 6.924 with a p-value of 0.00151. This indicates that small classes have a highly significant positive effect on reading scores (significant at the 1% level).
- Math Score: The estimated coefficient for small is 5.251 with a p-value of 0.143. Since the p-value is greater than 0.05, the effect of small classes on math scores is not statistically significant.
- Difference: Yes, there is a clear difference. Reducing class size significantly improves reading performance but does not provide statistically significant evidence of improving math performance.

(c)

```
star_ra <- subset(df, small == 0)

reg_c <- lm(totalscore ~ aide, data = star_ra)
summary(reg_c)

##
## Call:
## lm(formula = totalscore ~ aide, data = star_ra)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -191.748  -50.442   -5.442   40.558  312.558
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.559  257.529  <2e-16 ***
## aide         4.306      4.994   0.862   0.389
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.23 on 835 degrees of freedom
## Multiple R-squared:  0.0008898, Adjusted R-squared: -0.0003068
## F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

Interpretation of Estimates $\text{TOTALSCORE}_i = 916.442 + 4.306 \times \text{AIDE}_i$

- Intercept ($\hat{\gamma}_1 = 916.442$): The estimated average total aptitude score for students in regular-sized classes without a teacher aide is 916.44.
- Coefficient for AIDE ($\hat{\gamma}_2 = 4.306$): Students in regular-sized classes with a full-time teacher aide score, on average, 4.306 points higher than those without an aide.
- Conclusion: The p-value for the aide coefficient is 0.389, which is much greater than the 0.05 significance level, we fail to reject the null hypothesis. We conclude that adding a teacher aide to a regular-sized classroom does not have a statistically significant effect on students' overall learning.

(d)

```
reg_d1 <- lm(readscore ~ aide, data = star_ra)
summary(reg_d1)

##
## Call:
## lm(formula = readscore ~ aide, data = star_ra)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -74.665 -21.536  -2.536   15.464  194.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.460  296.341  <2e-16 ***
## aide         2.871       2.049   1.401    0.161
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 29.64 on 835 degrees of freedom
## Multiple R-squared:  0.002347,    Adjusted R-squared:  0.001152
## F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615

reg_d2 <- lm(mathscore ~ aide, data = star_ra)
summary(reg_d2)

##
## Call:
## lm(formula = mathscore ~ aide, data = star_ra)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -146.212  -31.212   -5.777   29.223  142.223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.355  205.464  <2e-16 ***
## aide         1.435       3.304   0.434    0.664
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.79 on 835 degrees of freedom
## Multiple R-squared:  0.0002258, Adjusted R-squared:  -0.0009715
## F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

Observations and Differences:

Reading Score: The estimated coefficient for aide is 2.871 with a p-value of 0.161 in a regular-sized class.

Math Score: The estimated coefficient for aide is 1.435 with a p-value of 0.664.

Difference: Both p-values are well above 0.05. Therefore, adding a teacher aide does not significantly improve reading scores nor math scores. There are no major differences observed here, the presence of an aide remains statistically insignificant across all subjects.

Problem 2.25

```
str(cex5_small)
```

```
## 'data.frame':  1200 obs. of  10 variables:
## $ advanced: int  0 0 0 0 0 1 0 0 0 0 ...
## $ alcbev  : num  11.1 0 10 11.1 0 ...
## $ appar   : num  0 6.22 0 29.44 22.44 ...
## $ college : int  0 0 1 1 1 0 0 0 0 0 ...
## $ entert  : num  43.4 17.6 5 20 56 ...
## $ food    : num  9.63 94.22 38.52 173.33 173.33 ...
## $ foodaway: num  47.6 77 39.8 0 108.3 ...
## $ health  : num  58.22 3.33 33.33 130.59 135.11 ...
## $ income  : num  123.7 59.7 81 133.2 81 ...
## $ smsa    : int  1 1 1 1 0 1 1 1 1 1 ...
## - attr(*, "var.labels")= chr [1:10] "=1 if highest educ in HH Masters or Professional/PhD degree" "
```

(a)

```
summary(cex5_small$foodaway)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   12.04   32.55   49.27   67.50  1179.00
```

```
cat("Mean:  ", mean(cex5_small$foodaway, na.rm = TRUE), "\n")
```

```
## Mean:      49.27085
```

```
cat("Median: ", median(cex5_small$foodaway, na.rm = TRUE), "\n")
```

```
## Median:    32.555
```



```
cat("Q25:    ", quantile(cex5_small$foodaway, 0.25, na.rm = TRUE), "\n")
```

```
## Q25:      12.04
```

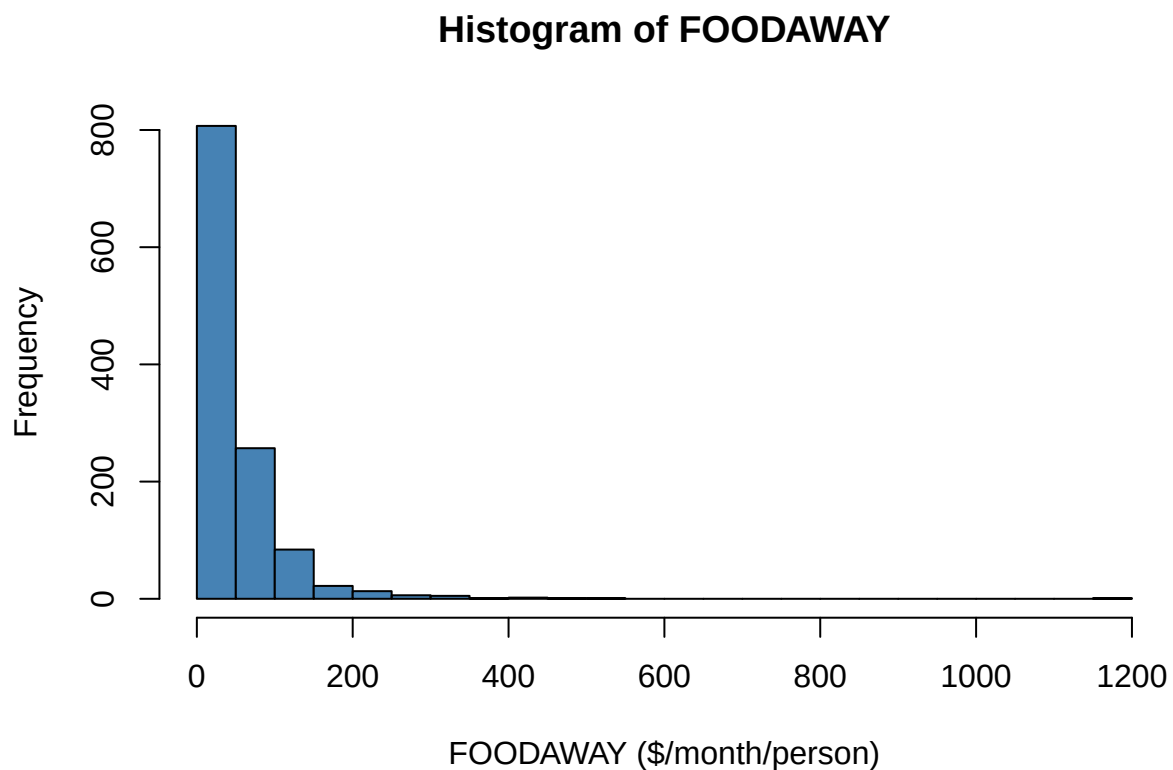
```
cat("Q75:    ", quantile(cex5_small$foodaway, 0.75, na.rm = TRUE), "\n")
```

```
## Q75:      67.5025
```

```
cat("Std Dev:", sd(cex5_small$foodaway, na.rm = TRUE), "\n")
```

```
## Std Dev: 65.28361
```

```
hist(cex5_small$foodaway, breaks = 30, col = "steelblue",  
     main = "Histogram of FOODAWAY",  
     xlab = "FOODAWAY ($/month/person)")
```



```
## (b)
```

```
# Advanced degree  
adv <- cex5_small$foodaway[as.logical(cex5_small$advanced)]  
cat("=== Advanced Degree ===\n")
```

```
## === Advanced Degree ===
```

```

cat("N:      ", length(adv), "\n")

## N:      257

cat("Mean:   ", mean(adv, na.rm = TRUE), "\n")

## Mean:    73.15494

cat("Median:", median(adv, na.rm = TRUE), "\n\n")

## Median:  48.15

# College degree (excluding advanced)
col <- cex5_small$foodaway[as.logical(cex5_small$college) & !as.logical(cex5_small$advanced)]
cat("=== College Degree (no advanced) ===\n")

## === College Degree (no advanced) ===

cat("N:      ", length(col), "\n")

## N:      369

cat("Mean:   ", mean(col, na.rm = TRUE), "\n")

## Mean:    48.59718

cat("Median:", median(col, na.rm = TRUE), "\n\n")

## Median:  36.11

# No college or advanced degree
nocol <- cex5_small$foodaway[!as.logical(cex5_small$college) & !as.logical(cex5_small$advanced)]
cat("=== No College/Advanced Degree ===\n")

## === No College/Advanced Degree ===

cat("N:      ", length(nocol), "\n")

## N:      574

cat("Mean:   ", mean(nocol, na.rm = TRUE), "\n")

## Mean:    39.01017

```

```
cat("Median:", median(nocol, na.rm = TRUE), "\n")
```

```
## Median: 26.02
```

(c)

```
cex5_small$lnfoodaway <- ifelse(cex5_small$foodaway > 0,  
                                log(cex5_small$foodaway), NA)
```

```
cat("FOODAWAY observations:    ", length(cex5_small$foodaway), "\n")
```

```
## FOODAWAY observations:    1200
```

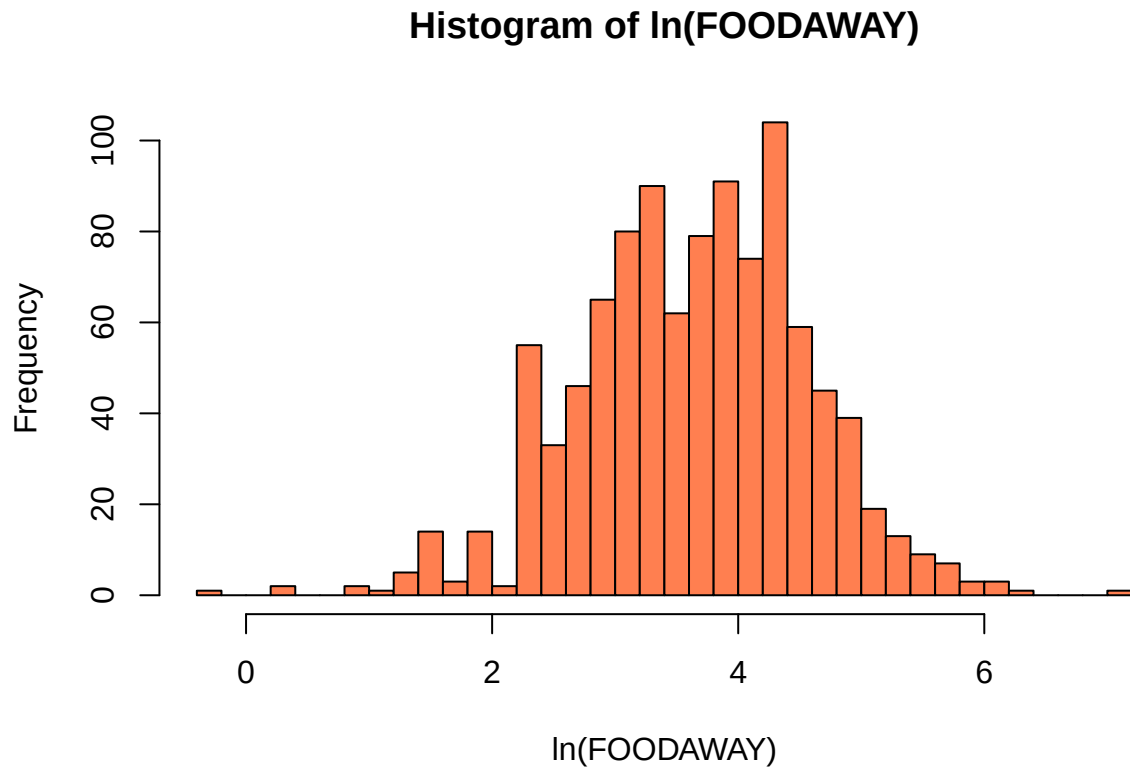
```
cat("ln(FOODAWAY) valid observations:", sum(!is.na(cex5_small$lnfoodaway)), "\n")
```

```
## ln(FOODAWAY) valid observations: 1022
```

```
summary(cex5_small$lnfoodaway)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's  
## -0.3011  3.0759   3.6865   3.6508  4.2797   7.0724     178
```

```
hist(cex5_small$lnfoodaway[!is.na(cex5_small$lnfoodaway)],  
     breaks = 30, col = "coral",  
     main = "Histogram of ln(FOODAWAY)",  
     xlab = "ln(FOODAWAY)")
```



The variable FOODAWAY has 1200 total observations. However, $\ln(\text{FOODAWAY})$ only has 1022 valid observations (with 178 NAs). This difference occurs because the logarithm function, $\ln(x)$, is only defined for strictly positive numbers ($x > 0$). In this dataset, there are 178 households that reported zero expenditure on food away from home ($\text{FOODAWAY} = 0$). When taking the natural log of 0, it results in an undefined value (represented as NA or missing in R).

(d)

```
reg_food <- lm(lnfoodaway ~ income, data = cex5_small)
summary(reg_food)
```

```
##
## Call:
## lm(formula = lnfoodaway ~ income, data = cex5_small)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6547 -0.5777  0.0530  0.5937  2.7000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.1293004  0.0565503   55.34  <2e-16 ***
## income       0.0069017  0.0006546   10.54  <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8761 on 1020 degrees of freedom
## (178 observations deleted due to missingness)
## Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
## F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

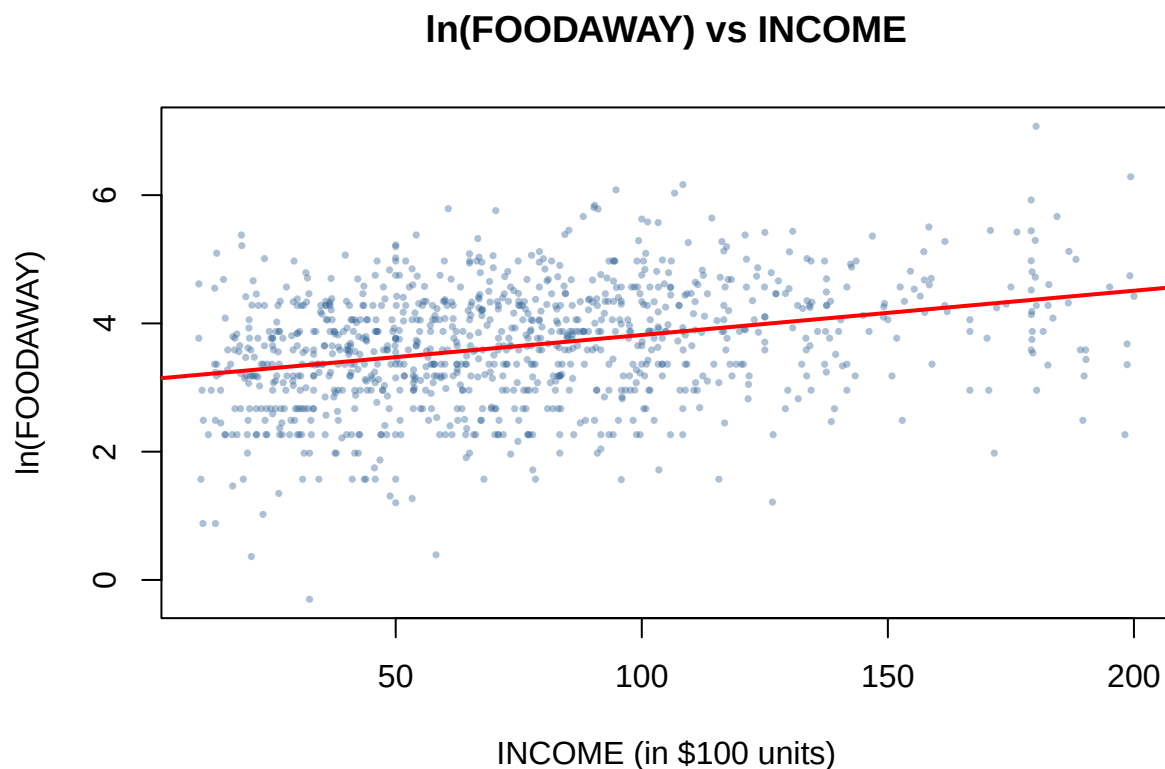
Interpretation of the estimated slope:

$$\widehat{\ln(\text{FOODAWAY})} = 3.1293 + 0.0069017 \times \text{INCOME}$$

- Estimated Slope ($\hat{\beta}_2 = 0.0069017$): Since the dependent variable is log-transformed ($\ln(Y)$) and the independent variable (X) is linear, this is a log-linear model. The coefficient interpretation is that for every 1 unit increase in INCOME (which is in \$100 units), the expenditure on food away from home (FOODAWAY) is estimated to increase by approximately 0.69%, holding all other factors constant.

(e)

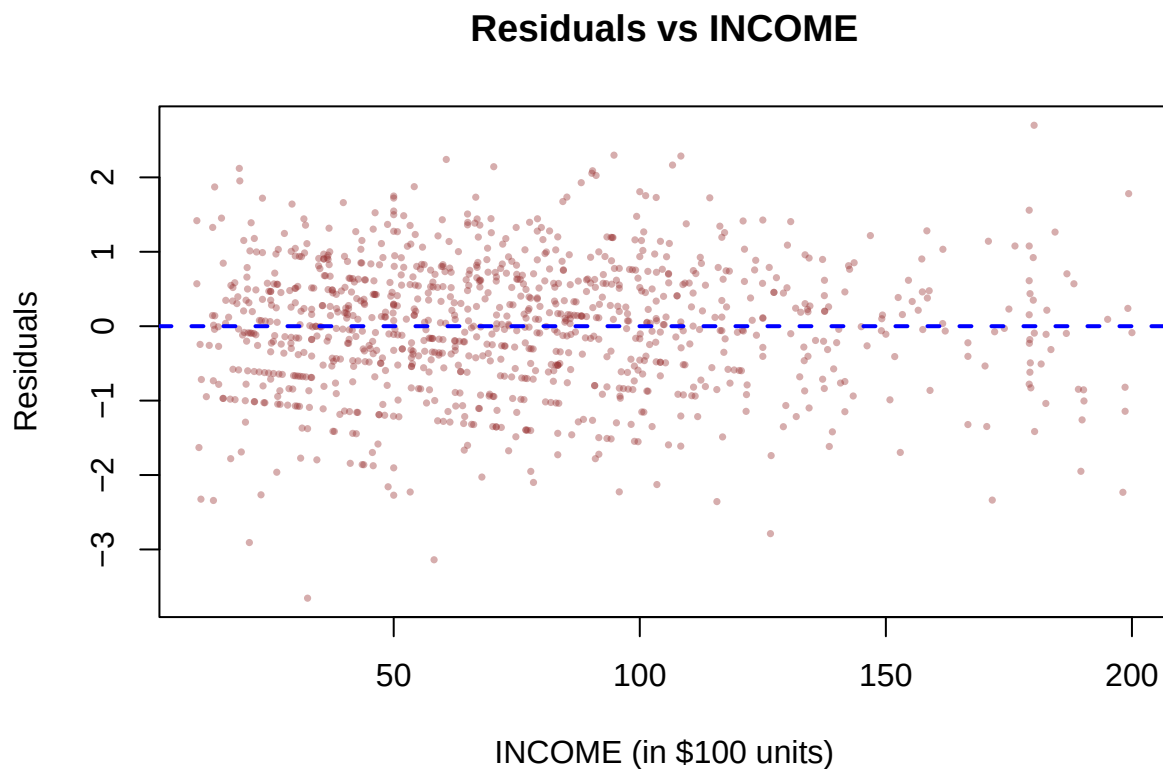
```
plot(cex5_small$income, cex5_small$lnfoodaway,
     pch = 16, cex = 0.5,
     col = rgb(0.2, 0.4, 0.6, 0.4),
     main = "ln(FOODAWAY) vs INCOME",
     xlab = "INCOME (in $100 units)",
     ylab = "ln(FOODAWAY)")
abline(reg_food, col = "red", lwd = 2)
```



(f)

```
residuals_f <- residuals(reg_food)
income_f <- cex5_small$income[!is.na(cex5_small$lnfoodaway)]

plot(income_f, residuals_f,
     pch = 16, cex = 0.5,
     col = rgb(0.6, 0.2, 0.2, 0.4),
     main = "Residuals vs INCOME",
     xlab = "INCOME (in $100 units)",
     ylab = "Residuals")
abline(h = 0, col = "blue", lwd = 2, lty = 2)
```



White Test for Heteroscedasticity

```
resid_sq <- residuals_f^2
white_reg <- lm(resid_sq ~ income_f + I(income_f^2))
summary(white_reg)
```

```
##
## Call:
## lm(formula = resid_sq ~ income_f + I(income_f^2))
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0248 -0.6893 -0.4059  0.1806 12.5562
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.256e-01  1.301e-01   7.117 2.08e-12 ***
## income_f      -4.690e-03  3.160e-03  -1.484   0.138
## I(income_f^2)  2.612e-05  1.654e-05   1.579   0.115
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.165 on 1019 degrees of freedom
## Multiple R-squared:  0.002455, Adjusted R-squared:  0.0004966
## F-statistic: 1.254 on 2 and 1019 DF, p-value: 0.2859
```

```
# White test statistics = n * R^2
n <- length(residuals_f)
r_sq <- summary(white_reg)$r.squared
white_stat <- n * r_sq
p_value <- 1 - pchisq(white_stat, df = 2)

cat("White Test Statistic (nR^2):", white_stat, "\n")
```

```
## White Test Statistic (nR^2): 2.50851
```

```
cat("Degrees of Freedom:      ", 2, "\n")
```

```
## Degrees of Freedom:      2
```

```
cat("P-value:                  ", p_value, "\n")
```

```
## P-value:                  0.2852883
```

Analysis of Residuals:

Visual inspection of the ‘Residuals vs INCOME’ plot reveals that the residuals are generally centered around the zero line. While the dispersion of residuals appears to widen at higher income levels (particularly when INCOME exceeds 100), this visual fanning effect is likely an artifact of data sparsity in the right tail rather than a true structural change in variance. This observation is formally supported by the White test ($nR^2 = 2.51$, $p = 0.285$), which fails to reject the null hypothesis of homoscedasticity at the 5% significance level. Consequently, there is no statistically significant evidence of heteroscedasticity, allowing us to reasonably conclude that the error terms exhibit constant variance.